Faster Algebraic Shifting

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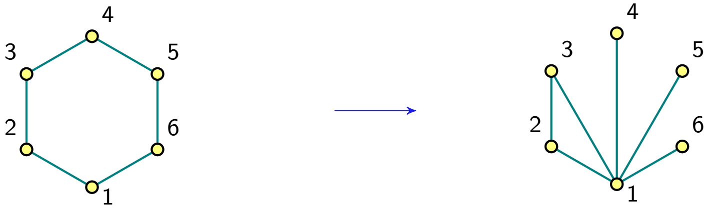
Algebraic Shifting

Definition A k-uniform hypergraph is a collection $S \subseteq \binom{[n]}{k}$.
S is shifted if for any $\sigma \in S$, $\sigma|_i^j \in S$ for $i \in \sigma$ and $j < i$.



S $\Delta(S)$

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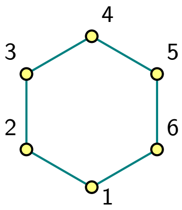
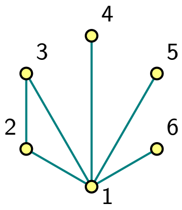


S

$\longrightarrow$



$\Delta(S)$

Theorem Kalai '84, Björner-Kalai '88 Given Σ a simplicial complex and a field $\mathbb{F}$, there exists a shifted simplicial complex $\Delta^{\mathbb{F}}(\Sigma)$ with the same f-vector and Betti numbers as Σ.

Theorem DV, Joswig, Lenzen '25 There exists a fast in practice Las Vegas algorithm for determining $\Delta^{\mathbb{F}}(S)$.

```
julia> using Oscar

julia> S = uniform_hypergraph([[1, 2], [2, 3], [3, 4], [4, 1]])
UniformHypergraph(4, 2, [[1, 2], [1, 4], [2, 3], [3, 4]])

julia> exterior_shift(GF(2), S)
UniformHypergraph(4, 2, [[1, 2], [1, 3], [1, 4], [2, 3]])
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Problem Kalai '02 Does there exist a deterministic or Las Vegas polynomial time algorithm for determining $\Delta(S)$?

Theorem Keehn-Nevo '24+ There exists a deterministic polynomial time algorithm for computing $\Delta^{\mathbb{Q}}(\Sigma)$ when Σ is a triangulation of either the torus, the Klein bottle or the real projective plane.

Exterior Algebraic Shift

Definition Let $\mathbb{E} \supset \mathbb{F}$ be a field extension and let $g \in \mathrm{GL}(n, \mathbb{E})$.

- The kth compound matrix $g^{\wedge k}$ is the $\binom{n}{k} \times \binom{n}{k}$-matrix of k-minors of g.
- $g^{\wedge S}$ is the $|S| \times \binom{n}{k}$-submatrix of $g^{\wedge k}$ with rows indexed by S.

Let $\sigma_i \in S$, $|S| = m$ and $\tau_i \in \binom{[n]}{k}$ such that

$$g^{\wedge S} = \begin{pmatrix} \det g_{\sigma_1 \tau_1} & \det g_{\sigma_1 \tau_2} & \cdots & \det g_{\sigma_m \tau_{\binom{n}{k}}} \\ \vdots & \vdots & \vdots & \vdots \\ \det g_{\sigma_m \tau_1} & \det g_{\sigma_m \tau_2} & \cdots & \det g_{\sigma_m \tau_{\binom{n}{k}}} \end{pmatrix} .$$

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The (exterior) algebraic shift is,

$$\Delta_g^{\mathbb{F}}(S) := \left\{ \sigma \in \binom{[n]}{k} : \sigma \text{ indexes a column with a pivot of } g^{\wedge S} \right\}.$$

for a "generic enough" g.

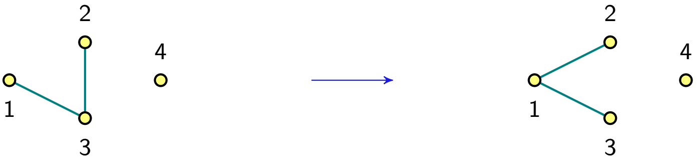
Small Example

For $g = (x_{ij})_{1\le i,j\le 4}$ $S = \{\{1,2\},\{2,3\},\} \subset \binom{[4]}{2}$, $\mathbb{F} = \mathrm{GF}(2)$.

$$g^{\wedge S} = \begin{pmatrix} x_{11}x_{22}+x_{12}x_{21} & x_{11}x_{23}+x_{13}x_{21} & x_{11}x_{24}+x_{14}x_{21} & x_{12}x_{23}+x_{22}x_{13} & x_{12}x_{24}+x_{14}x_{22} & x_{13}x_{24}+x_{14}x_{23} \\ x_{21}x_{32}+x_{22}x_{31} & x_{21}x_{33}+x_{23}x_{31} & x_{21}x_{34}+x_{24}x_{31} & x_{22}x_{33}+x_{23}x_{32} & x_{22}x_{34}+x_{24}x_{32} & x_{23}x_{34}+x_{24}x_{33} \end{pmatrix} .$$

Which row reduces to,

$$\begin{pmatrix} x_{11}x_{22}+x_{12}x_{21} & x_{11}x_{23}+x_{13}x_{21} & x_{11}x_{24}+x_{14}x_{21} & x_{12}x_{23}+x_{13}x_{22} & x_{12}x_{24}+x_{14}x_{22} & x_{13}x_{24}+x_{14}x_{23} \\ 0 & \begin{array}{c} x_{11}x_{21}x_{22}x_{33} \\ +x_{11}x_{21}x_{23}x_{32} \\ +x_{12}x_{21}^2x_{33} \\ +x_{13}x_{21}^2x_{32} \\ +x_{12}x_{21}x_{23}x_{31} \\ +x_{13}x_{21}x_{22}x_{31} \end{array} & \begin{array}{c} x_{11}x_{21}x_{22}x_{34} \\ +x_{11}x_{21}x_{24}x_{32} \\ +x_{12}x_{21}^2x_{34} \\ +x_{14}x_{21}^2x_{32} \\ +x_{12}x_{21}x_{24}x_{31} \\ +x_{14}x_{21}x_{22}x_{31} \end{array} & \begin{array}{c} x_{11}x_{22}^2x_{33} \\ +x_{11}x_{22}x_{23}x_{32} \\ +x_{12}x_{21}x_{22}x_{33} \\ +x_{13}x_{21}x_{22}x_{32} \\ +x_{12}x_{22}x_{23}x_{31} \\ +x_{13}x_{22}^2x_{31} \end{array} & \begin{array}{c} x_{11}x_{22}^2x_{34} \\ +x_{11}x_{22}x_{24}x_{32} \\ +x_{12}x_{21}x_{22}x_{34} \\ +x_{14}x_{21}x_{22}x_{32} \\ +x_{12}x_{22}x_{24}x_{31} \\ +x_{14}x_{22}^2x_{31} \end{array} & \begin{array}{c} x_{11}x_{22}x_{23}x_{34} \\ +x_{11}x_{22}x_{24}x_{33} \\ +x_{12}x_{21}x_{23}x_{34} \\ +x_{12}x_{21}x_{24}x_{33} \\ +x_{13}x_{21}x_{24}x_{32} \\ +x_{14}x_{21}x_{23}x_{32} \\ +x_{13}x_{22}x_{24}x_{31} \\ +x_{14}x_{22}x_{23}x_{31} \end{array} \end{pmatrix} .$$



LU decomposition

Lemma $\Delta_{gb}(S) = \Delta_g(S)$ for any upper triangular b and any simplicial complex S.

LU decomposition For $g \in \mathrm{GL}(n, \mathbb{E})$ with non zero leading principal minors, there exists ℓ lower triangular, u upper triangular such that

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Example

$K = \{\{1,2\},\{2,3\},\{4\}\}$, and $\ell^{\wedge S}$ reduces to

$$\begin{pmatrix} x_{13}x_{24}+x_{14}x_{23} & x_{12}x_{24}+x_{14} & x_{24} & x_{12}x_{23}+x_{13} & x_{23} & 1 \\ 0 & \begin{matrix} x_{12}x_{23}x_{24}x_{34}+x_{12}x_{24}^2 \\ +x_{13}x_{24}x_{34}+x_{14}x_{24} \end{matrix} & \begin{matrix} x_{23}x_{24}x_{34} \\ +x_{24}^2 \end{matrix} & \begin{matrix} x_{12}x_{23}^2x_{34}+x_{12}x_{23}x_{24} \\ +x_{13}x_{23}x_{34}+x_{14}x_{23} \end{matrix} & \begin{matrix} x_{23}^2x_{34} \\ +x_{23}x_{24} \end{matrix} & \begin{matrix} x_{23}x_{34} \\ +x_{24} \end{matrix} \end{pmatrix},$$

Monte Carlo Algorithm

- Randomized algorithm.
- Current Standard for computing $\Delta^{\mathbb{F}}(S)$.
- Pick u by sampling from $\mathbb{F}$.
- Computing $\Delta^{\mathbb{F}}_u(S)$ is fast.
- Successful with high probability in characteristic 0.
- Not the case for finite fields.

Verification Algorithm (Las Vegas)

Algorithm Verify u computes the algebraic shift of S

Input: $u \in \mathbb{E}^{n\times n}$, S a k-uniform hypergraph.
Output: `true` if $\Delta^{\mathbb{F}}(S) = \Delta^{\mathbb{F}}_u(S)$; otherwise `false`

$S' \leftarrow \Delta_u(S)$
$\sigma_{\max} \leftarrow \max_{\text{lex}} S'$
$T \leftarrow \{\tau \in \binom{[n]}{k} : \tau <_{\text{lex}} \sigma_{\max}\}$
if $|S'| = |T| + 1$ **then**
  return `true`
end if
$m \leftarrow$ row echelon form of $\ell^{\wedge S}_{*\,T}$
if columns of pivots of $m = S' \setminus \{\sigma_{\max}\}$ **then**
  return `true`
end if
return `false`

$$\ell^{\wedge S} = \begin{pmatrix} | & | & \cdots & | & | & \cdots \\ c_1 & c_2 & \cdots & c_{|T|} & \sigma_{\max} & \cdots \\ | & | & \cdots & | & | & \cdots \end{pmatrix}$$

Simplicial Complexes

Definition An abstract simplicial complex Σ, is a collection of non empty sets of $[n] := \{1, 2, \ldots, n\}$, such that for any $\tau, \sigma \subset [n]$, if $\tau \subset \sigma \in K$ then $\tau \in \Sigma$. change lower case k

- $\sigma \in \Sigma$ a k-simplex or a face, $\dim(\sigma) = k + 1$.
- f-vector number of k-faces for each k.

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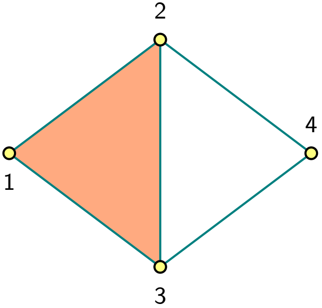

The f-vector is $(4, 5, 1)$

Simplicial Homology

Definition Given a field $\mathbb{F}$, and a simplicial complex Σ we define the kth simplicial homology to be the vector space

$$H_k^{\mathbb{F}}(\Sigma) := k\text{-cycles} \Big/ k+1\text{-boundaries}.$$

The kth <u>Betti number</u> of Σ is the dimension of $H_k^{\mathbb{F}}(\Sigma)$.

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The kth Betti number of Σ is the dimension of $H_k^{\mathbb{F}}(\Sigma)$.

The Betti numbers are $(1, 1, 0)$

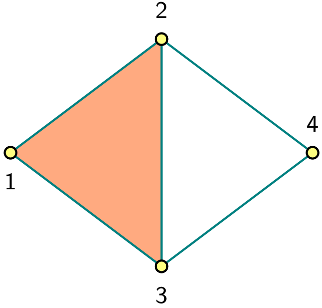

Table of timings

					$\mathbb{Q}$			GF(2)			GF(4)			
n	o	g	i	#2-faces	time	#trials		time	#trials		time	#trials		...
		⋮				⋮			⋮			⋮		
7	✗	1	2	12	0.1	1	*	6.5	14		6.9	6		...
7	✓	0	5	10	0.3	1		1.9	3		2.5	1		...
7	✓	1	1	14	2.7	1		578.7	2		554.0	1		...
8	✗	1	4	14	2.4	1		–	>100	*	11.9	5		...
8	✗	1	5	14	0.1	1	*	8.7	26		9.6	2		...
8	✗	1	6	14	0.1	1	*	10.6	75		11.7	2		...
8	✗	1	7	14	0.1	1	*	3.3	26		4.3	2		...
8	✗	2	1	16	0.1	1	*	1338.8	25		1350.0	2		...
8	✗	2	2	16	0.1	1	*	oot	oot		oot	oot		...
8	✓	0	1	12	1.1	1		4.6	84		5.5	2		...
8	✓	0	2	12	1.8	1		–	>100		8.1	2		...
8	✓	1	1	16	3.6	1		578.1	26		571.1	2		...
8	✓	1	3	16	27.0	1		oot	oot		oot	oot		...
8	✓	1	7	16	oom	oom		oot	oot		oot	oot		...
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋱

Table of timings

					$\mathbb{Q}$			GF(2)			GF(4)		
n	o	g	i	#2-faces	time	#trials		time	#trials		time	#trials	...
		$\vdots$				$\vdots$			$\vdots$			$\vdots$	
7	✗	1	2	12	0.1	1	*	6.5	14		6.9	6	...
7	✓	0	5	10	0.3	1		1.9	3		2.5	1	...
7	✓	1	1	14	2.7	1		578.7	2		554.0	1	...
8	✗	1	4	14	2.4	1		–	>100	*	11.9	5	...
8	✗	1	5	14	0.1	1	*	8.7	26		9.6	2	...
8	✗	1	6	14	0.1	1	*	10.6	75		11.7	2	...
8	✗	1	7	14	0.1	1	*	3.3	26		4.3	2	...
8	✗	2	1	16	0.1	1	*	1338.8	25		1350.0	2	...
8	✗	2	2	16	0.1	1	*	oot	oot		oot	oot	...
8	✓	0	1	12	1.1	1		4.6	84		5.5	2	...
8	✓	0	2	12	1.8	1		–	>100		8.1	2	...
8	✓	1	1	16	3.6	1		578.1	26		571.1	2	...
8	✓	1	3	16	27.0	1		oot	oot		oot	oot	...
8	✓	1	7	16	oom	oom		oot	oot		oot	oot	...
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$

Thank you!

Table of sizes

					$\mathbb{Q}$			GF(2)			
n	o	g	i	#2-faces	#terms	max-degree		#terms	max-degree		...
		⋮				⋮			⋮		
7	✗	1	2	12	n/a	n/a	*	1982	15		...
7	✓	0	5	10	1314	12		1182	12		...
7	✓	1	1	14	530	15		524	15		...
8	✗	1	4	14	13445	20		n/a	n/a	*	...
8	✗	1	5	14	n/a	n/a	*	7952	18		...
8	✗	1	6	14	n/a	n/a	*	8256	19		...
8	✗	1	7	14	n/a	n/a	*	2188	17		...
8	✗	2	1	16	n/a	n/a	*	53046	22		...
8	✗	2	2	16	n/a	n/a	*	oot	oot		...
8	✓	0	1	12	9300	15		8344	15		...
8	✓	0	2	12	9142	15		8288	15		...
8	✓	1	1	16	5179	20		4818	20		...
8	✓	1	3	16	20293	18		oot	oot		...
8	✓	1	7	16	oom	oom		oot	oot		...
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋱